



Jinmaitong ameliorates diabetes-induced peripheral neuropathy in rats through Wnt/ β -catenin signaling pathway

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ABSTRACT

Ethnopharmacological relevance: Jinmaitong (JMT) is a prescription of Traditional Chinese Medicine, which is composed of ten herbal drugs and two animal drugs. It has long been used for the treatment of diabetic peripheral neuropathy (DPN).

Aim of study: Wnt/ β -catenin pathway is considered as an essential and direct driver of myelinogenesis. This study aims to evaluate the protective effect of JMT against DPN dynamically during a 16-weeks' treatment, and to investigate the underlying mechanism in which the Wnt/ β -catenin pathway is involved.

Materials and methods: Diabetic model was induced by single intraperitoneal injection of Streptozotocin (STZ) using male Sprague-Dawley rats. The model rats were divided into five groups and administrated with JMT at three doses (0.437, 0.875, and 1.75 g/kg per day), neurotrophin (positive drug, 2.67 NU/kg per day), and placebo (deionized water), respectively, for continuous 8 weeks (n = 9–10), 12 weeks (n = 8–10), or 16 weeks (n = 7–9). Meanwhile, rats in control group were administrated with placebo (n = 10 for 8 weeks, n = 9 for 12 and 16 weeks, respectively). Blood glucose and body weight were monitored every four weeks. Mechanical allodynia was assessed using mechanical withdrawal threshold (MWT) test. The morphological change of sciatic nerves were observed by transmission electron microscope (TEM) and hematoxylin and eosin (HE) stain. The mRNA and protein levels of targeted genes were evaluated by quantitative real time-PCR and western blot, respectively. Myelin protein zero (MPZ) and mediators involved in Wnt/ β -catenin pathway, such as β -catenin, glycogen synthase kinase 3 β (GSK-3 β), and WNT inhibitory factor-1 (WIF-1), were compared among different groups after treatment of 8, 12, and 16 weeks, respectively.

Results: The mechanical allodynia and peripheral nerve morphology were degenerated in DPN rats over time, and notably improved after JMT-treatment of 12 and 16 weeks. The decreased MPZ level in DPN rats were also significantly amended by JMT. More importantly, we found that the suppressed Wnt/ β -catenin pathway in sciatic nerves of DPN rats was overtly up-regulated by JMT in a time-dependent manner. Among the three doses, JMT at the middle dose showed the best effect.

Conclusions: JMT effectively ameliorated diabetes-induced peripheral neuropathy, which was mediated by the activation of Wnt/ β -catenin signaling pathway. This study provided new perspective to understand the neuro-protective mechanism of JMT.

1. Introduction

Diabetic peripheral neuropathy (DPN) is a common complication of diabetes, which affected more than half of the diabetic patients during disease progression, leading to neuropathic pain and increasing risk of

ulcer and amputation in long period (Pop-Busui et al., 2017). Demyelination is the most typical feature in pathological development of DPN, which is closely related to nerve dysfunction and decreased nerve regeneration (Zenker et al., 2013). Therefore, remyelination is a key point in promoting peripheral nerve function recovery and relieving the DPN symptoms. Although increased aldose reductase activity (Yagihashi

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Abbreviations

DPN	diabetic peripheral neuropathy
NTP	neurotrophin
JMT	Jinmaitong
IGF-1	insulin-like growth factor 1
GSK-3 β	glycogen synthase kinase 3 β
WIF-1	Wnt inhibitory factor-1
UHPLC	ultra-high performance liquid chromatography
MS	mass spectrometry
SD	Sprague-Dawley
STZ	streptozocin

MWT	mechanical pain threshold
TEM	transmission electron microscope
HE	hematoxylin and eosin
MPZ	myelin protein zero
AOD	average integrated optical density
qPCR	quantitative real-time PCR
GAPDH	glyceraldehyde-3-phosphate dehydrogenase
p- β -catenin	phosphorylated β -catenin
p-GSK-3 β	phosphorylated GSK-3 β
SEM	standard error of mean
ANOVA	one-way analysis of variance

et al., 2001), inflammation stress (Schmeichel et al., 2003), impaired neurotrophic support (Lupachyk S et al., 2013), and some other factors have been proposed in DPN development, the pathogenesis of this diabetic complication was still not fully clarified yet. As a result, no efficacious therapy is available other than intensive glycaemic control and symptomatic treatment in clinical practice (Javed et al., 2015). Neurotrophin (NTP), a non-protein extract from inflamed rabbit skin inoculated with vaccinia virus, has been used in clinical to alleviate neuropathic pain in DPN patients (Ishikawa et al., 2015). However, it is not effective enough in long-term therapeutic procedure and possible to cause side effects on digestive and nervous system (Sakai et al., 2018). More effective medical therapy of DPN is still in strong demand.

Jinmaitong (JMT) is a compound prescription which has been used in the treatment of DPN for several decades. It is composed of 12 kinds of Traditional Chinese Medicines, including 10 herbal drug materials and two animal drug materials. Our randomized double-blind clinical trial showed that JMT significantly accelerated nerve conduction and improve the clinical symptoms of DPN with favorable security (Liang et al., 1999; Wu et al., 2012). Besides, our series of studies have indicated that the potential mechanism of JMT in attenuating DNP included affecting red blood cell aldolase reductase activity, improving expression of nerve growth factor (Shi et al., 2013a), insulin-like growth factor 1 (IGF-1), and IGF-1 receptor (Song et al., 2019), as well as suppressing thioredoxin-interacting protein and Nod-like receptor protein 3 inflammasome activation (Sun et al., 2019). However, Traditional Chinese Medicine is characterized by multi-targets in treatment of diseases, which could be also applied to JMT in DPN treatment. The underlying neuroprotective mechanism of JMT still needs further study.

It is worth noting that the disorder of certain signaling pathway was also triggered in demyelination and DPN development. For instance, the Wnt/ β -catenin signaling pathway, also known as the canonical Wnt signaling pathway, has been proved to play a key role in neurogenesis and the fate of neural progenitors (Gao et al., 2010). Researchers found that this pathway was also an essential and direct driver of myelin gene expression and myelinogenesis (Tawk et al., 2011; Ortega et al., 2013). Thus, development of new neuroprotective drugs targeting Wnt/ β -catenin pathway becomes a potential therapeutic intervention of DPN (Serafino et al., 2020). However, the dynamic changes on this signaling pathway during the progression of DPN and whether JMT acts on this pathway remains unknown.

In the present study, the neuroprotective effect of JMT was confirmed on DPN rats. Then the expression of pivotal signaling involving Wnt/ β -catenin pathway, such as β -catenin, glycogen synthase kinase 3 β (GSK-3 β), and WNT inhibitory factor-1 (WIF-1), were determined from the sciatic nerve of rats during DPN progression and JMT treatment of 8, 12, and 16 weeks. These results interpreted the neuroprotective mechanism of JMT from a new perspective, which laid foundation for the application of JMT in DPN treatment.

2. Materials and methods

2.1. Preparation of JMT

Jing-Mai-Tong (JMT) is composed of the well mixed fine powder of 12 drug materials, including 10 kinds of herbal drugs (Semen Cuscutae, Fructus Ligustri Lucidi, Herba Ecliptae, Herba Prunella Vulgaris, Semen Litchi, Ramulus Cinnamomi, Rhizoma Corydalis, Semen Persicae, Semen Cassiae, and Radix et Rhizoma Asari) and two kinds of animal drugs (scorpio and hirudo) in a fixed ratio of 10: 10: 10: 10: 30: 3: 10: 10: 10: 30: 3: 3 as our previously report (Song et al., 2019). The detailed information of each drug material were given in Table 1 and Supplementary Fig. 1. All of the crude drugs were purchased from Tong Ren Tang Lit. Corp (Beijing, China) and authenticated by Prof. X.C. Liang (Peking Union Medical College Hospital, Beijing, China). The voucher specimens (No. JMT-15A – JMT-15L) were deposited at the specimen room of Department of Traditional Chinese Medicine, Peking Union Medical College Hospital, Beijing, China. JMT was suspended in deionized water before intragastric administration.

Table 1

Detailed information of the crude drug materials in JMT.

Voucher number	Drug name	Biological origin	Chinese name
JMT-15A	Semen Cuscutae	the seeds of <i>Cuscuta chinensis</i> Lam.	Tu-Si-Zi
JMT-15B	Fructus Ligustri Lucidi	the seeds of <i>Ligustrum lucidum</i> W.T. Aiton	Nv-Zhen-Zi
JMT-15C	Herba Ecliptae	the herb of <i>Eclipta prostrata</i> (L.) L.	Mo-Han-Lian
JMT-15D	Herba Prunella Vulgaris	the herb of <i>Prunella vulgaris</i> L.	Xia-Ku-Cao
JMT-15E	scorpio	the whole body of <i>Buthus martensii</i> Karsch	Li-Zhi-He
JMT-15F	Semen Litchi	the seeds of <i>Litchi chinensis</i> Sonn.	Quan-Xie
JMT-15G	Ramulus Cinnamomi	the tender stem of <i>Cinnamomum cassia</i> (L.) J. Presl	Gui-Zhi
JMT-15H	Rhizoma Corydalis	the rhizoma of <i>Corydalis yanhusuo</i> (Y.H.Chou & Chun C.Hsu) W.T. Wang ex Z.Y.Su & C.Y.Wu	Yan-Hu-Suo
JMT-15I	Semen Persicae	the seeds of <i>Prunus persica</i> (L.) Batsch	Tao-Ren
JMT-15J	Senmen Cassiae	the seeds of <i>Cassia obtusifolia</i> L.	Jue-Ming-Zi
JMT-15K	hirudo	the whole body of <i>Whitmania pigra</i> Whitman	Shui-Zhi
JMT-15L	Radix et Rhizoma Asari	the radix and rhizoma of <i>Asarum heterotropoides</i> F.Schmidt.	Xi-Xin

Note: The species in column of biological origins have been validated via <http://www.plantsoftheworldonline.org/> for botanic drug materials and via Catalogue of Life China (2019) (<http://www.sp2000.org.cn/>) for animal drug materials.

2.2. Chemical analysis of JMT

A 100-mg aliquot of JMT powder was extracted with 10 mL of ethanol under ultrasonic bath for 20 min. The extracting solution was filtered with a 0.22- μ m membrane before injection. Sample analysis was performed on an ultra-high performance liquid chromatography (UHPLC) coupled with TripleTOFTM 5600 mass spectrometer (Sciex, USA). Separation was achieved on an ACQUITY HSS T3 column (100 mm $\times$ 2.1 mm, 1.7 μ m) (Waters, USA). Mobile phase A and mobile phase B were acetonitrile and water containing 0.1% formic acid, respectively. Gradient elution program was applied (0–3 min, 5–20% A; 3–10 min, 20–40% A; 10–20 min, 40–75% A; 20–25 min, 75–95% A; 25–30 min, 95% A) with flow rate at 0.30 mL/min. The column temperature was maintained at 40 °C. MS data was acquired in ESI (–) ion mode with a wide mass range (m/z 50 to 1500). The curtain gas, gas 1 and gas 2 of the mass spectrometer were set at 0.24, 0.27 and 0.27 MPa, respectively. The ion source temperature was 450 °C, and the spray voltage was 4500 V.

Following this approach, the chemical composition of JMT was profiled and shown in Supplementary Figure 1M. A total of 59 compounds were putatively identified by comparing their high-resolution MS data with literatures, all of which had been identified from the equal-mixed methanol extract and water extract of JMT in our previous study (Song et al., 2019). The detailed information of the 59 compounds were listed in Supplementary Table 1.

2.3. Animals and treatments

In animal experiment, all possible attempts were made to alleviate the suffering of the rats according to the Principles of laboratory animal care (<http://grants1.nih.gov/grants/olaw/references/phspol.htm>), which was approved by the Experimental Animal Ethics Committee of Peking Union Medical College Hospital (application No. XHDW-2015-0025). Male Sprague-Dawley (SD) rats at five-weeks old were obtained from Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China) and housed in conditions of specific pathogen-free grade under a light/dark cycle (light on at 8:00 a.m. and off at 8:00 p.m.). Standard diet and distilled water were freely available. All rats were acclimatized for seven days prior to study.

Neurotrophin (NTP) was taken as positive drug in this study and purchased from Nippon Zoki Pharmaceutical Co Ltd (Osaka, Japan, registration certificate No. S20140086, lot No. 16009). Streptozocin (STZ) was from Sigma-Aldrich (St Louis, MO, USA, catalog No. S0130-1G, lot No. WXB6558V). Diabetic rats were induced by intraperitoneal injection of STZ in 0.9% sodium chloride at dose of 55 mg/kg after a 12 h-fasting. Blood glucose was measured from the tail tip using the Accu-Chek Active Blood Glucose Meter (Roche Diabetes Care Inc., Ireland). Rats with blood glucose >16.7 mmol/L after 72 h since STZ injection were considered as diabetic rats and then randomly divided into STZ group, STZ + JMT-L group, and STZ + JMT-M group, STZ + JMT-H group, and STZ + NTP group ($n = 30$ at the starting time point). Additionally, 30 male SD rats of matched age and weight were intraperitoneally injected with 0.9% sodium citrate of the same volume and employed as control group.

Among the six groups, rats in STZ + JMT-L group, STZ + JMT-M group, and STZ + JMT-H group were orally administrated with JMT at dose of 0.4375 g/kg/d (five-fold of the dose for a human adult), 0.875 g/kg/d, and 1.75 g/kg/d, respectively. The dosage was calculated based on the well-mixed JMT powder. Rats in STZ + NTP group were orally administrated with the positive drug, NTP, at dose of 2.67 nU/kg/d. Beside, rats in control group and STZ group were administrated with distilled water of equal volume (5 mL). Treatments were conducted once per day and for consecutive 8 weeks ($n = 9$ –10), 12 weeks ($n = 8$ –10), or 16 weeks ($n = 7$ –9).

2.4. Mechanical pain threshold (MWT) assessment

Weight and blood glucose were measured every four weeks for each rat. At the last day of 8w, 12 w and 16w, MWT assessment was performed using Von Frey Pain Measurement Instrument (IITC Life Science Inc., USA). The experimental details for MWT assessment were same as that described in our recent publication (Sun et al., 2019).

2.5. Morphological observation of sciatic nerve

The morphological changes of sciatic nerves in different groups were compared by histopathological examination and transmission electron microscope (TEM) observation. After MWT assessment, rats were anesthetized by intraperitoneally injection of chloral hydrate (300 mg/kg) and urethane (ethyl carbamate, 0.75 g/kg) after treatment of 8 w, 12 w and 16 w, respectively. Sciatic nerves were isolated bilaterally. The left sciatic nerve was cut into two segments. One segment (3–4 mm length) was fixed by 4% PFA, embedded in paraffin, and cut into both cross and longitudinal sections at a thickness of 4 μ m. Sections were stained by Hematoxylin and Eosin (HE), and imaged by ultra-compact image capture device (Aperio CS2, Leica, Germany). The other segment (1–2 mm length) was fixed in 2.5% glutaraldehyde at 4 °C overnight. After further fixation for 1 h in 1% osmic acid, sample was rinsed three times with 0.1M PBS and dehydrated by graded ethanol. After treatment with absolute ethanol and embedding solution (1:1.5) at room temperature for 1 h, the sample was processed with absolute ethanol and embedding solution (1:3) also at room temperature for 2 h. After soaking in pure resin overnight, the sample was processed in pure embedding solution at 37 °C for 4 h. Finally, specimens were incubated at 45 °C for 12 h and 60 °C for 48 h, cut at 70 nm using an ultra-microtome, and then stained with 2% uranyl acetate-lead citrate. The myelin sheath was observed by TEM at 12 000 $\times$ and 4000 $\times$ magnification (TEM-1400 plus, JEOL, Tokyo, Japan). To assess the axonal myelination, g-ratio (axon diameter/myelinated fiber diameter) was calculated using a plug-in (<http://gratio.efil.de/>) of Image J software, which allowed semiautomatic analysis of individual myelinated axons (Goebbels, 2010). A minimum of 50 axons were randomly selected for g-ratio analysis for each rat.

2.6. Immunohistochemistry

The expression of myelin protein zero (MPZ) in sciatic nerves were detected by immunohistochemical method. The tissues were dewaxed, rehydrated, repaired of hot antigens with citric acid (pH 6.0), and then treated with 3% H₂O₂ solution. The tissues were incubated with MPZ antibody (ab31851, Abcam, Cambridge, UK) at 4 °C overnight and subsequently incubated with secondary antibody at room temperature for 30 min. After that, the sections were stained with diaminobenzidine solution, counterstained with hematoxylin, and dehydrated. The mark protein was noticed under Leica DM3000 microscope and the image acquisition system (Leica Microsystems, Wetzlar, Germany) under united conditions and measured by average integrated optical density (AOD, the integrated optical density/area of interesting).

2.7. Quantitative real-time PCR

Sciatic nerve samples were homogenized on ice. The total RNA was extracted with Trizol reagent according to the manufacturer's instructions. Reverse transcription of RNA was conducted using first strand cDNA synthesis kit (TaKaRa, Dalian, China). The quantitative real-time PCR (qPCR) were performed with equal amounts of cDNA using the ChamQTM Universal SYBR qPCR Master Mix (Vazyme Biotech Co., Ltd., Nanjing, China) in a LineGene real-time PCR detection system (Applied Biosystems, Thermo Fisher Scientific Co., Ltd., MA, USA). The forward and reverse primer sequences (5'–>3') for MPZ, β -catenin, GSK-3 β , WIF-1, and GAPDH were listed in Supplementary Table 2. The reaction program was set as follows: Pre-incubation at 95 °C for 5 min;

amplification at 95 °C for 10 s, 59 °C for 60 s, 40 cycles; separation at 95 °C for 10 s, and 60 °C for 60 s. Standard curves were generated for target genes after amplification, and then normalized against endogenous reference gene GAPDH. Samples were run in duplicates and relative expression of the target genes was determined by $2^{-\Delta\Delta Ct}$ method.

2.8. Western blot

The expression of MPZ, β -catenin, phosphorylated β -catenin (p- β -catenin), GSK-3 β , phosphorylated GSK-3 β (p-GSK-3 β), and WIF-1 in sciatic nerves were detected by western blot. Sciatic nerve samples were homogenized with RIPA lysis buffer on ice. The protein concentration was determined using BCA protein assay kit (Beyotime, Shanghai, China). Proteins were separated using 10% SDS-PAGE and then transferred onto a nitrocellulose membranes (Pall Gelman, MI, USA). After blocking with 5% skim milk for 1 h at room temperature, primary antibodies [anti- β -catenin (ab32572), anti- β -catenin (phospho S33 + S37) (ab32572), anti-GSK-3 β (ab93926), anti-GSK-3 β (phospho Ser 9) (ab107166), anti-WIF1 (ab155101), and anti-MPZ (ab31851), all from Abcam, Cambridge, UK] were added at 4 °C for overnight incubation. Secondary antibody labeled with HRP (ab205718, Abcam, Cambridge, UK) were then added and stained at 37 °C for 1 h. Image J software was used to analyze the protein bands, using β -actin as the internal reference protein for normalization.

2.9. Statistical analysis

GraphPad Prism 8.0 software (La Jolla, CA, USA) was applied for statistical analysis. Data were presented as mean $\pm$ standard error of mean (SEM). Data from two groups were compared using unpaired two-tailed Student's *t*-test, while data from multiple groups were compared using one-way analysis of variance (ANOVA) followed by Tukey's post hoc test. $P < 0.05$ was considered statistically significant.

3. Results and discussions

3.1. Effect of JMT on blood glucose level and body weight

We first investigated the neuroprotective effect of JMT at three dosages during a 16 weeks' therapy. The blood glucose level and body weight of each rat were measured every four weeks. As shown in Fig. 1A, the blood glucose levels of rats in STZ group, STZ + JMT-L/M/H groups, and STZ + NTP group were significantly higher than control group throughout the 16 weeks ($P < 0.01$). Nevertheless, when compared with STZ group, the blood glucose levels showed slight decrease since the 8th week in STZ + JMT-M and STZ + NTP groups, and since the 12th week in

STZ + JMT-L/H groups ($P < 0.05$). As the animal model employed in this study was type 1 diabetic rat model induced by STZ, which is in absolute absence of insulin, the potential hypoglycemic effect of JMT observed here might not be insufficient to demonstrate its clinical significance. As shown in Fig. 1B, the body weight curve of control group showed continued and significant growth during the 16 weeks, and much higher than STZ model group and JMT-treatment groups ($P < 0.01$). Although the body weight of JMT-M/H and NTP treated groups were higher than the STZ group since the 8th week ($P < 0.05$), there was no overt increasing trend over time in all diabetic rats since the 4th week.

3.2. Effect of JMT on MWT

The hindpaw MWT of each rat was measured to evaluate the degree of DPN and the therapeutic effect of JMT. As shown in Fig. 2, the MWT of STZ group was significantly lower than control group since the 8th week ($P < 0.01$), indicating the DPN rat model was established successfully. It's worth noting that the decreasing trends of MWT caused by peripheral neuropathy were remarkably alleviated by JMT at three different doses after treatment of 8 weeks ($P < 0.05$), 12 weeks ($P < 0.01$), and 16 weeks ($P < 0.01$). Among them, JMT at middle dose showed the best effect and was even better than NTP on improving MWT.

3.3. Effect of JMT on morphology of the sciatic nerve

To further determine the protective effect of JMT against DPN, gross changes of sciatic nerve histology were observed by light microscopy using rats treated for 16 weeks. The cross section of HE staining (Fig. 3A) showed that the diameter of sciatic nerve fibers was larger, and the nerve fibers were more compact in normal control group when compared with the STZ model group. The axons of myelinated nerve fibers of in normal control group were wrapped around a myelin sheath of uniform thickness, while the nerve fibers in STZ model group were different in size. In longitudinal section (Fig. 3B), the nerve fibers of normal rats were arranged orderly and closely. In contrast, the structure of nerve fibers in STZ model group was disordered, with loose arrangement and partial fibers fracture. The nerve fiber in STZ model group exhibited obvious demyelination changes and axonal shrinkage from cross section. Visible signs of vacuolar-like defects and separated lamellar spaces were evident in STZ model group from both cross and longitudinal sections. However, the morphological injuries of sciatic nerve in DPN rats were obviously improved by JMT and NTP treatment. Among three doses of JMT, the middle dose exhibited better effect than the other two doses. As shown in Fig. 3A, the size of nerve fiber in these treatment groups were larger than STZ group. Vacuolar-like

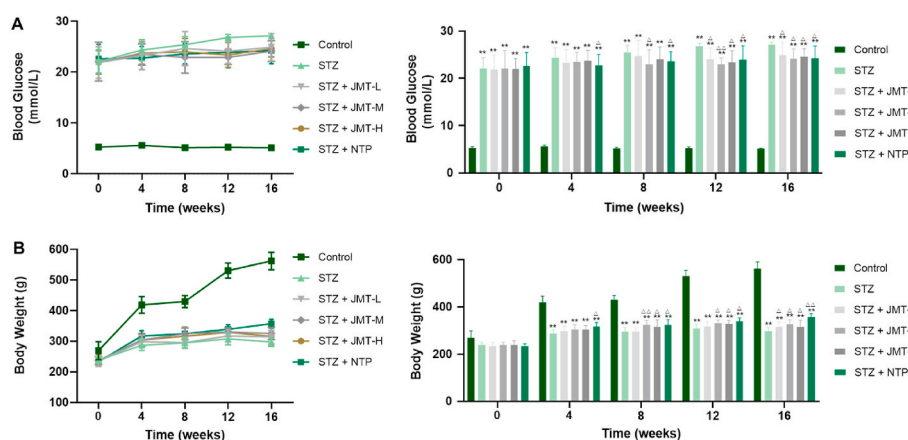


Fig. 1. Effect of JMT on blood glucose level (A) and body weight (B). Data were shown in mean $\pm$ SEM ($n = 7-10$). ** $P < 0.01$ vs. Control group; $\Delta P < 0.05$ vs. STZ group; $\triangle\triangle P < 0.01$ vs. STZ group.

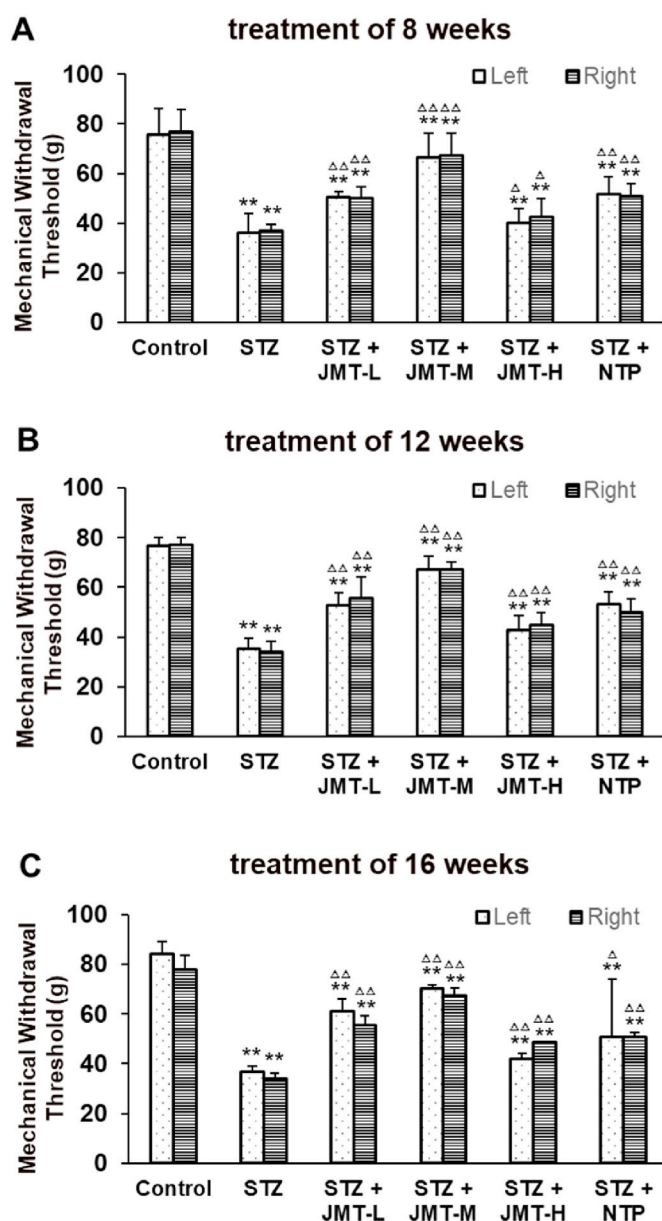


Fig. 2. Effect of JMT on the left- and right-paw mechanical withdrawal threshold (MWT) in different groups after 8 weeks (A), 12 weeks (B), and 16 weeks (C) of treatment. Data were shown in mean $\pm$ SEM ($n = 7-10$); ** $P < 0.01$ vs. Control group; $\Delta P < 0.05$ vs. STZ group; $\Delta\Delta P < 0.01$ vs. STZ group.

degeneration in myelin sheath and demyelination were significantly reduced in JMT-M, JMT-H and NTP treated group. Axonal injury were also prevented in JMT-M, JMT-H, and NTP treated groups.

Moreover, the ultrastructural damage of sciatic nerve was further assessed by TEM, while the progression of DPN and the effect of JMT were observed over time using rats had been treated for 8, 12, and 16 weeks, respectively (Fig. 4). At the 8th week, the axons of myelinated fibers from the control group were surrounded by a typical myelin sheath with a clear, uniform, and lamellar structure. In contrast, myelin sheath deformation was observed in sciatic nerve of STZ group at the same time, indicating the DPN model was successfully established 8 weeks after STZ injection. The prominent ultrastructural alterations of the myelin sheath in DPN rats were vacuolization and lamellar separation, which were aggravated from the 8th week to the 16th week. Accordingly, there was a decrease in g-ratio of DPN rats since the 12th week ($P < 0.05$, versus normal controls) (Fig. 4C). Besides, slight

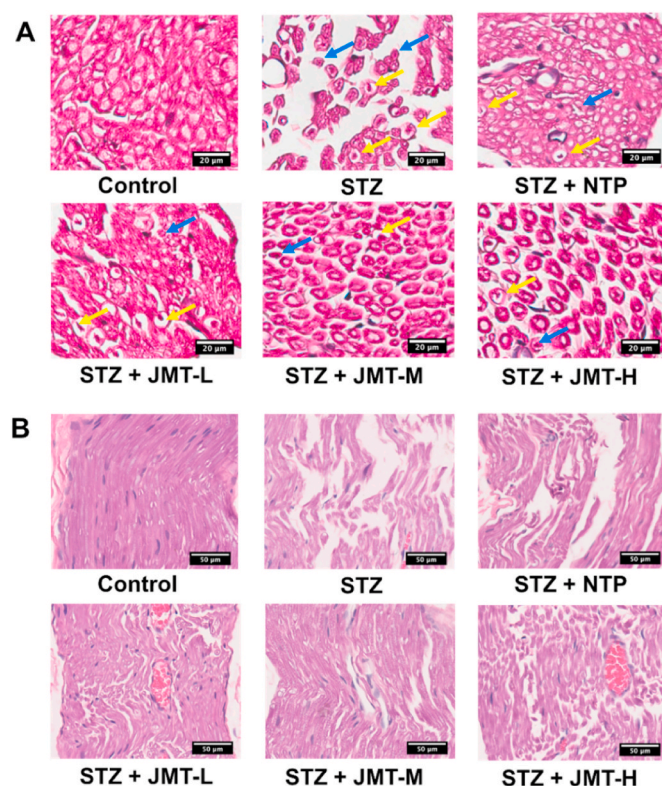


Fig. 3. Representative HE staining photomicrographs of cross-section (A) and longitudinal section (B) from Control, STZ, STZ + JMT-L, STZ + JMT-M, STZ + JMT-H, and STZ + NTP groups. Sciatic nerves of STZ group exhibited obvious vacuolar-like defects (yellow arrows), demyelination changes and axonal shrinkage (blue arrows). Magnification, 400 $\times$; scar bars, 20 μ m for A and 50 μ m for B. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

lamellar separation and axonal shrinkage were also observed in normal control group at the 16th week (Fig. 4B). These ultrastructure damages were ameliorated in varying degrees by JMT treatment at different dosages. JMT at middle dose showed a better therapeutic effect, especially after treatment of 12 and 16 weeks, which was in accordance with the results of MWT test and HE staining. The vacuolization and lamellar separation of myelin sheath were renovated visibly. Morphological improvements of the sciatic nerve were also evidenced by the increase of g-ratio in JMT-M and JMT-H treated groups at the 16th week ($P < 0.05$, versus untreated DPN controls).

3.4. JMT up-regulated the MPZ level in sciatic nerves

MPZ constitutes a major component of peripheral myelin. The mRNA and protein levels of MPZ in sciatic nerve was measured after treatment of 8, 12, and 16 weeks by qPCR and western blot, respectively. The immunohistochemical analysis of MPZ was also conducted at the 16th week (Fig. 5). As shown in Fig. 6 and Fig. 7, the MPZ expression at both mRNA and protein levels trended downward from the 8th to the 16th week in both control group and DPN model group, indicating the decline of myelin regeneration over time. On the other hand, the MPZ expression were significantly decreased in STZ group compared to control group ($P < 0.01$) at all the three time points. This decreasing trend was remarkably reserved by JMT-L/M/H at the 8th, 12th, and 16th week ($P < 0.01$), among which JMT-M group exhibited the highest MPZ expression at both mRNA and protein level. The MPZ level in JMT-treated groups was most close to control group at the 16th week, indicating a longer treatment period came with better efficacy (Figs. 6A, 7B and C). Among the three doses, JMT at middle dose showed the most

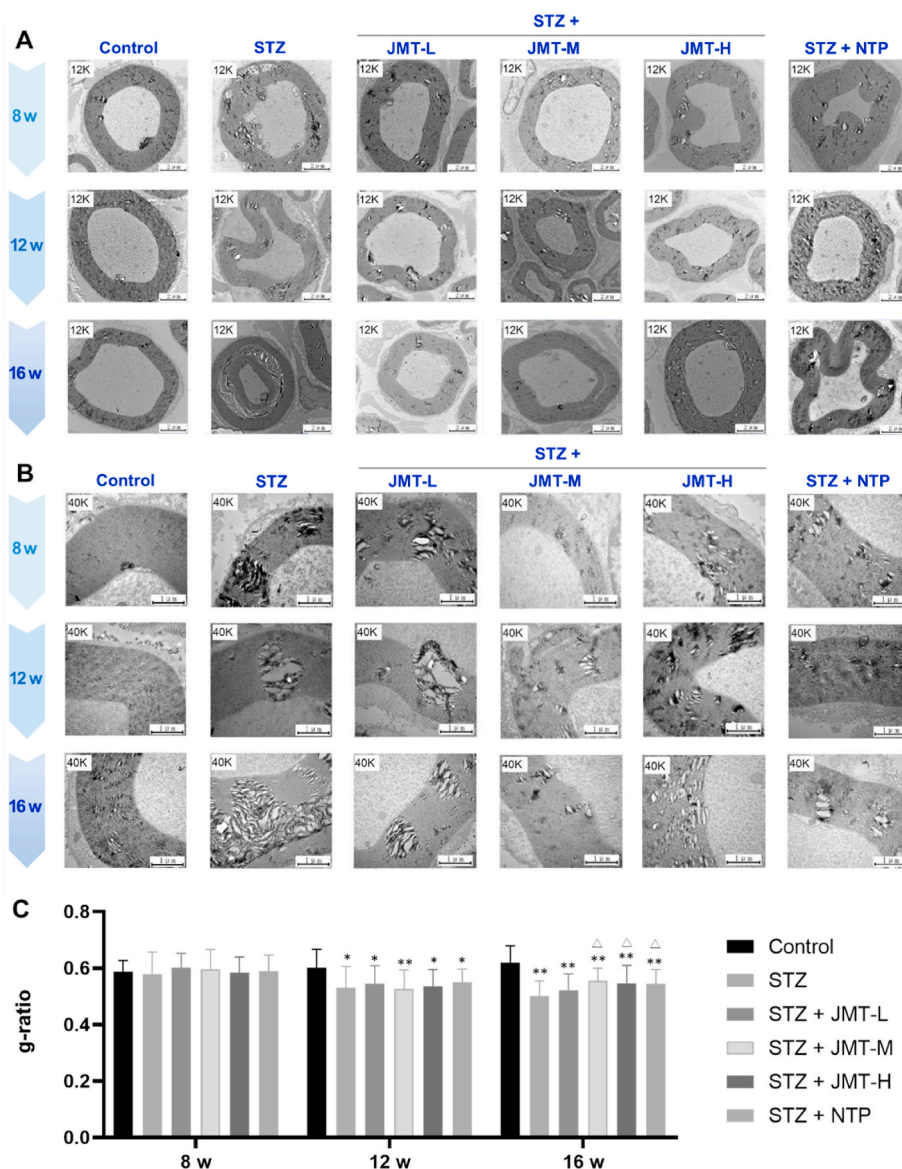


Fig. 4. Ultrastructure of sciatic nerve assessed by TEM. Representative sciatic nerve images of the cross-sections from rats in the Control, STZ, STZ + JMT-L, STZ + JMT-M, STZ + JMT-H, and STZ + NTP groups at magnifications of 12000 $\times$ (A) and 40000 $\times$ (B), respectively. Quantification of myelin thickness by g-ratio analysis (C). Data were shown in mean $\pm$ SEM (n = 3 rats); *P < 0.05 vs. Control group; **P < 0.01 vs. Control group; Δ P < 0.05 vs. STZ group.

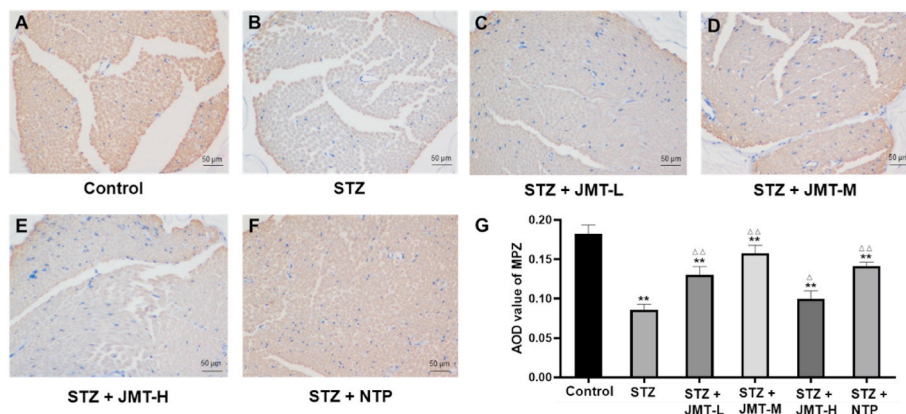


Fig. 5. Immunohistochemical analysis of MPZ after treatment of 16 weeks. Cross sections of sciatic nerves in Control (A), STZ (B), STZ + JMT-L (C), STZ + JMT-M (D), STZ + JMT-H (E), and STZ + NTP (F) groups were observed at a magnification of 400 $\times$, respectively. G, AOD of MPZ in different groups; data were shown in mean $\pm$ SEM (n = 3); **P < 0.01 vs. Control group; Δ P < 0.05 vs. STZ group; $\Delta\Delta$ P < 0.01 vs. STZ group.

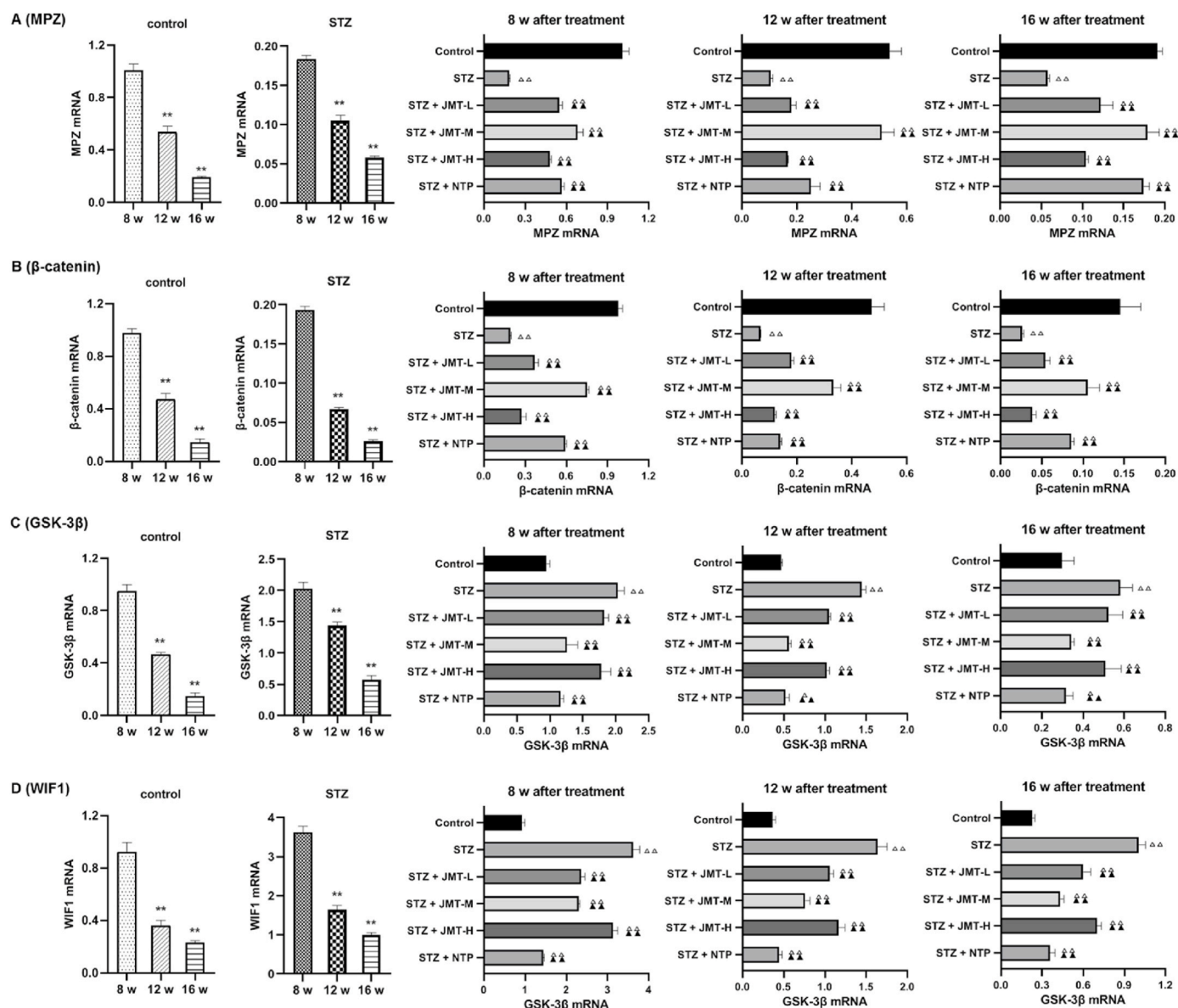


Fig. 6. The mRNA level of MPZ (A), β -catenin (B), GSK-3 β (C), and WIF-1 (D) after different administration periods among the six groups, respectively. Data were normalized to the normal control group at the 8th week and shown in mean $\pm$ SEM ($n = 3$); * $P < 0.05$, ** $P < 0.01$, vs. sub-group of 8 weeks; $\Delta P < 0.05$, $\Delta\Delta P < 0.01$, vs. control group; $\Delta P < 0.05$, $\Delta\Delta P < 0.01$, vs. STZ group.

remarkable effect on up-regulation of the MPZ level. The result was also supported by immunohistochemical analysis, for the AOD of MPZ in JMT-M group was higher than other treatment groups at the 16th week (Fig. 5).

3.5. Effect of JMT on Wnt/ β -catenin signaling pathway

To explore whether Wnt/ β -catenin signaling pathway was involved in the progress of DPN and the pharmacological mechanism of JMT, we further assessed the dynamic variations of the core regulator (β -catenin) and two inhibitors (GSK-3 β and WIF-1) of the Wnt/ β -catenin signaling pathway among different groups at both mRNA and protein level.

First of all, we found that the mRNA level of β -catenin and the protein level of total and phosphorylated β -catenin were overtly decreased within control and STZ groups over time, respectively ($P < 0.01$, Figs. 6B, 7D and E). Then the β -catenin levels were compared among different groups at the 8th, 12th, and 16th week, respectively. The β -catenin level in STZ group was significantly lower than control group

at both mRNA and protein levels all the time, while p- β -catenin showed an opposite trend ($P < 0.01$), indicating the inhibition of Wnt/ β -catenin signaling pathway. The decrease of β -catenin and increase of p- β -catenin in DPN rats were remarkably reversed by JMT-L/M/H, as well as the positive drug NTP ($P < 0.01$). All these findings indicated that JMT could up-regulated the Wnt/ β -catenin signaling pathway in sciatic nerves of DPN rats. Among the JMT-treated groups, JMT-M group showed the most significant anoxoaction on this pathway.

The expression of both WIF-1 and GSK-3 β showed obvious slippage over time at mRNA and protein levels (Fig. 6C and D, Fig. 7F, G, and 7H). When compared to the control group, the WIF-1 expression was much higher in STZ group ($P < 0.01$), indicating the Wnt/ β -catenin signaling pathway was inhibited in sciatic nerve of DPN rats. The abnormal elevation of WIF-1 was restrained by JMT-L/M/H significantly since the 8th week at mRNA level ($P < 0.01$) and since the 12th week at protein level ($P < 0.05$). As for GSK-3 β , analogous variation trend was observed at mRNA level. JMT-L/M/H remarkably down-regulated the over-expressed mRNA of GSK-3 β in DPN rats ($P < 0.01$). Nevertheless,

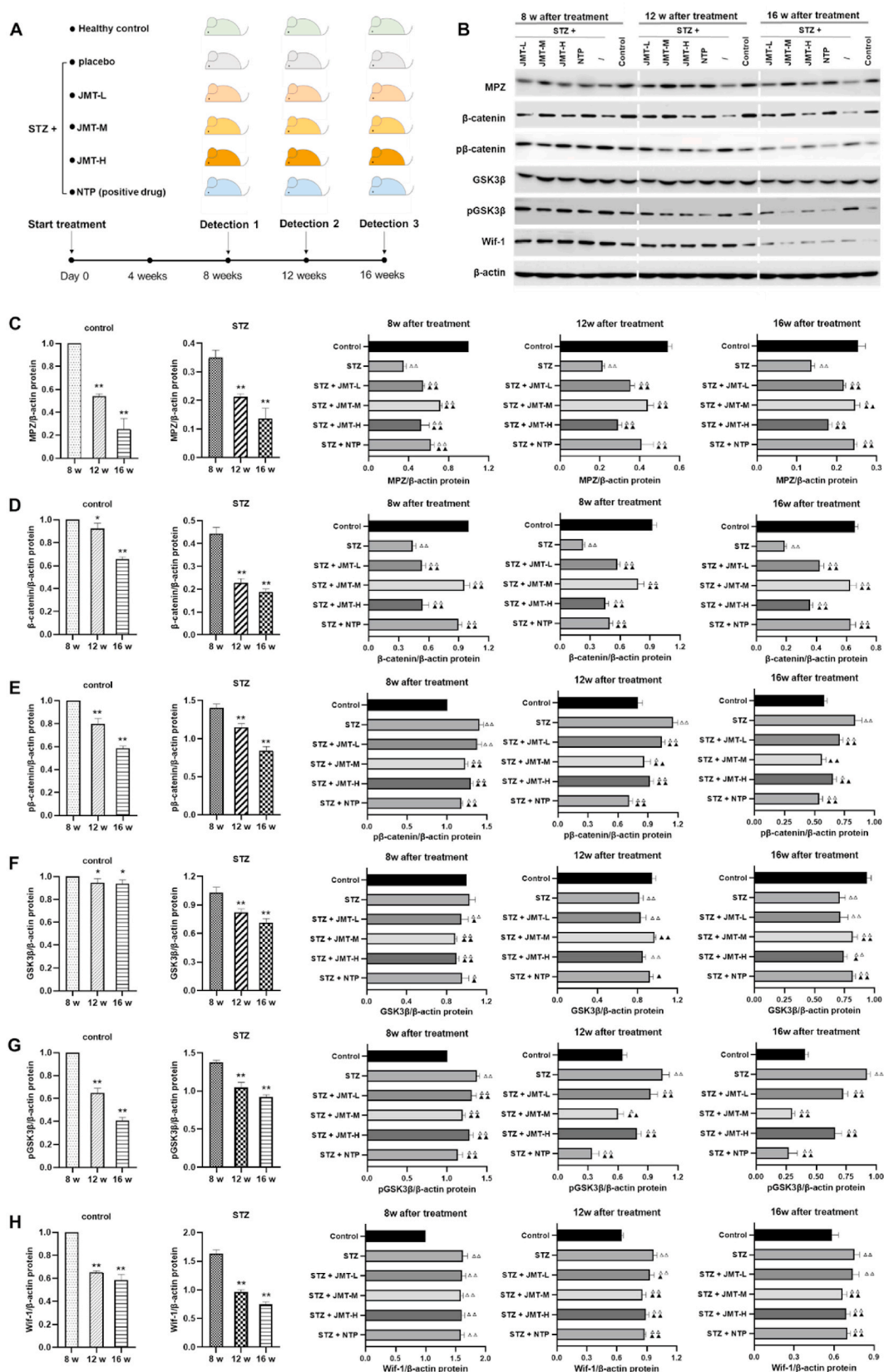


Fig. 7. The effect of JMT on expression of MPZ and five proteins involving the canonical Wnt/ β -catenin signaling pathway. **A**, animal grouping and the detection time points. **B**, Representative Western blot images. **C–H**, densitometric analysis of MPZ/ β -actin, β -catenin/ β -actin, p β -catenin/ β -actin, GSK-3 β / β -actin, p-GSK-3 β / β -actin, and WIF-1/ β -actin, respectively. Data were normalized to the normal control group at the 8th week and shown in mean $\pm$ SEM ($n = 3$); * $P < 0.05$, ** $P < 0.01$, vs. sub-group of 8 weeks; $\Delta P < 0.05$, $\Delta\Delta P < 0.01$, vs. control group; $\blacktriangle P < 0.05$, $\blacktriangle\blacktriangle P < 0.01$, vs. STZ group.

when we further evaluated the protein expression of total and phosphorylated GSK-3 β , it was found that JMT treatment mainly effect GSK-3 β expression by reducing the up-regulation of the p-GSK-3 β in sciatic nerve of DPN rats ($P < 0.01$), while the total GSK-3 β showed a limited rise (within 10%) in JMT-treated groups since the 12th weeks.

As a whole, these results demonstrated that the Wnt/ β -catenin signaling pathway was inhibited in sciatic nerves of DPN rats, and activated by JMT in a time-dependent manner. Among the three doses, JMT-M showed the most significant effect among the three doses, which was in accordance with the histopathological results. Besides, the positive drug NTP also displayed a consistent effect with JMT on this pathway.

4. Discussion

In this study, typical symptoms of DPN had been apparent after 8 weeks of STZ-injection in type 1 diabetes rat model, which was confirmed by the ultrastructural changes of sciatic nerve under TEM (Fig. 4). JMT administration was applied since the hyperglycemic model was established, while the neuroprotective effect was firstly observed at the 8th week and maintained until the 16th week. The mechanical allodynia and pathologic changes were alleviated obviously, proving JMT as an effective therapy for peripheral neuropathy in diabetic rats.

MPZ is a kind of glycoprotein which only existed in the peripheral myelin sheath. It is vital to maintain the cohesion between adjacent myelin wraps and the normal structure of myelinated nerves (Garbay et al., 2000). MPZ gene was found to be up-regulated in the process of myelin formation and played an important role in promoting myelin regeneration (Umapathi et al., 2007). Recent research found that the recovery of MPZ lagged behind the functional recovery of peripheral nerves (Speck et al., 2019). In spite of this, myelin-specific proteins such as MPZ were still the most important indicator to identify novel myelinating drug candidates (Preston et al., 2019). Drugs specifically targeted myelin membrane and stimulated the MPZ production were promising in treatment of peripheral nerve injury (Ahmed et al., 2019; Li et al., 2019). Therefore the mRNA and protein levels of MPZ were also taken as a pharmacodynamics indicator of JMT in this study. We found that JMT could improve the MPZ expression in a time-related manner. The disparity of MTZ level between the JMT-treated group and normal control group was shrunk over time. The mRNA and protein levels of MPZ in DPN rats was closest to the normal rats at the 16th week.

Wnt/ β -catenin signaling pathway is critical involved in the determination of cell fate during embryogenesis and tissue renewal. It has been recognized that this pathway play a critical role in neurodevelopment and remyelination in both the central and peripheral nervous system (Huang et al., 2019). In peripheral nervous system, the activation of Wnt/ β -catenin signaling pathway increased the number of oligodendrocytes, as well as enhanced the MPZ expression in Schwann's cell, the peripheral nerve glial cell that surround the axons. Accordingly, the inhibition of this pathway suppressed the expression of MPZ and myelin basic protein (Fonte et al., 2005; Toledo et al., 2008). Studies in recent five years indicated that the Wnt/ β -catenin signaling pathway might be a potential therapeutic target for neurodegenerative diseases including DPN (Feng et al., 2015; Serafino et al., 2017) and diabetic cutaneous ulcers (Zhang et al., 2018). Some natural compounds from traditional medicines, such as ginsenoside Rg1 (Zhou et al., 2016) and ginkgolide B (Li et al., 2018), were found to display neuroprotective effect on neuropathological model through activating the Wnt/ β -catenin pathway. In our study, the dynamic variations of the Wnt/ β -catenin signalling pathway in the DPN pathologic process were monitored. Meanwhile, the effect of JMT on this pathway during a 16-weeks' therapy was evaluated for the first time.

Beta-catenin acts as the key signal transducer in Wnt/ β -catenin signaling, which determines the downstream cascades through its intracellular concentration and phosphorylation level (Tawk et al., 2011). The Wnt/ β -catenin signaling is activated when the extracellular

Wnt ligands is combined to LRP 5/6 and Frizzled protein families on cell surface, leading to the increase of β -catenin level in cytoplasm. The accumulated β -catenin transfers into nucleus and binds to the Tcf/Lef transcription factors, which eventually leads to the up-regulation of target genes (Ji et al., 2019). In fact, increased β -catenin level indicates the activation of Wnt/ β -catenin pathway (Voronkov et al., 2013). Oppositely, the Wnt/ β -catenin signaling is inactive when β -catenin forms a complex with the adenomatous polyposis coli protein (APC) and Dishevelled (Dsh)/Axin, and then phosphorylated by GSK-3 before it is eventually degraded (Fig. 8A). Recent study discovered that limited Dsh/Axin oligomerization determined the efficiency of signal transduction in this pathway (Kan et al., 2020).

Our present study revealed that the mRNA and protein expressions of β -catenin were significantly suppressed in the injured sciatic nerve of DPN rat. After JMT treatment of 8, 12 or 16 continuous weeks, significant increase of β -catenin at both mRNA and protein levels were observed in sciatic nerve, which was accompanied by the decrease of p- β -catenin (Fig. 8B). These results were in accordance with our previous study *in vitro*, in which we found that JMT promoted myelin sheath recovery by stimulating Schwann cell proliferation in high glucose medium, and enhanced the secretion of MPZ through up-regulating of β -catenin (Sun et al., 2017). Thus, both our *in vivo* and *in vitro* studies revealed the up-regulation of β -catenin and down-regulation of p- β -catenin after JMT treatment, indicating the activation of Wnt/ β -catenin signaling was, at least in part, involved in the neuroprotective effect of JMT against peripheral neuropathy.

To further validate the effect of JMT on this pathway, multi-proteins at upstream or downstream of the pathway still need to be determined. As the name implies, Wnt inhibitory factor-1 (WIF-1) acts as a Wnt antagonist. Besides, GSK-3 β is a multi-functional serine/threonine protein kinase, which regulates phosphorylation of β -catenin in cytoplasm and target it for degradation. GSK-3 β can be inactivated by phosphorylation at Ser9 residue (Lehtovirta et al., 2019). Therefore, WIF-1 and GSK-3 β are generally recognized as the inhibitor of this pathway. In sciatic nerve of DPN rats, the mRNA and protein level of WIF-1, as well as the mRNA level of GSK-3 β were all remarkably up-regulated, indicating the inhibition of the Wnt/ β -catenin pathway. The variation of these indicators were in accordance with β -catenin, the pivotal molecule in this pathway. Nevertheless, it is noteworthy that the protein level of GSK-3 β showed an opposite trend in our study. It is difficult to interpret this result, however we need to take into account that increasing evidences have proved that the Wnt signaling pathway is much more complex than previously thought. In recent years, the activation of Wnt/ β -catenin pathway has been found to occur in a GSK-3 independent manner (Devotta et al., 2018). Contrariwise, some cellular processes have been proved to be modulated by GSK-3 independent of Wnt/ β -catenin (Jope and Johnson, 2004). Recently, GSK-3 β was reported to be a join-point between the PI3K/AKT pathway and the Wnt/ β -catenin pathway (Vallée, 2020). From the above, we hypothesize that multiple pathways participate in DPN progress. Further studies are still in need to target more signaling networks involved in DPN using transcriptomics or proteomics technology, and to understand the cellular mechanism of JMT more comprehensively.

As a compound prescription, JMT contained multiple constituents with distinctive chemical structures and various bioactivities, some of which have been found to be able to alleviate DPN in our previous studies. The neuroprotective effect of quercetin, cinnamaldehyde, and hirudin, three representative compounds from three main drug materials of JMT (Semen Cuscutae, Ramulus Cinnamomi, and hirudo, respectively), have been validated. As a result, all of these compounds showed protective effect on rat dorsal root ganglion (DRG) neurons against high glucose-induced injury *in vitro* (Shi et al., 2013b; Yang et al., 2016; Liu et al., 2020). The apoptosis of DRG neurons induced by high glucose was found to be reduced significantly by the treatment with combination of the three active ingredients from JMT, while the activation of Nrf-2/HO-1 and the inhibition of NF- κ B were found to be

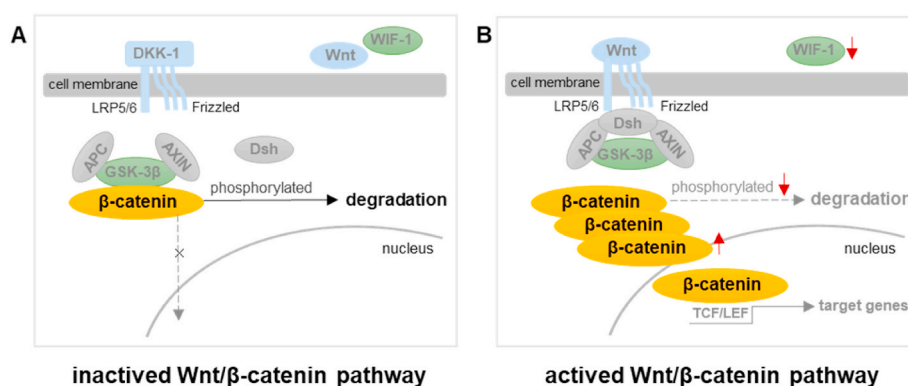


Fig. 8. The WNT/β-catenin signaling pathway. A, inactive status; B, active status. Red arrows indicated the effect of JMT (↑, up-regulation; ↓, down-regulation). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

involved in (Shi et al., 2017). Quercetin was also existed in another two herbs from JMT (*Eclipta prostrata* and *Prunella vulgaris*), and was able to attenuate thermal hyperalgesia on DPN rats (Anjaneyulu et al., 2003). In addition to quercetin, more flavonoids identified from JMT, such as hesperidin, luteolin, apigenin, and scutellarin (Table S1), have been reported to relieve experimental peripheral neuropathy obviously *in vivo* (Carballo-Villalobos et al., 2017; Li et al., 2015; Kim et al., 2019; Basu et al., 2020). All these compounds contributed to the pharmaceutical effect of JMT. In fact, among the 59 compounds identified by UHPLC/MS analysis, about a third were flavonoids and their glycosides, which were best known as strong antioxidants and attenuated different kinds of neuropathic pain at behavioral, physiological, and biochemical levels (Basu et al., 2020). Aside from these known active compounds, we believed that the effect of JMT against DPN was the comprehensive efforts of multiple components. Screening of novel neuroprotective components from JMT could be taken into account in future research.

In addition, three different doses of JMT were applied in this study. JMT of middle dose, which is ten-fold of the dose for human adult, showed the most significant alleviating effect on hyperpathia and the injured peripheral nerves. Since about six-fold of the dose for human adult is bioequivalent for rat, the dosage used in clinic still requires more consideration. As for reason why JMT at middle dose showed better effect than JMT at high dose, there are three main considerations. Firstly, JMT at high dose might cause some side effects on DPN rats, such as gastrointestinal discomfort which lead to poorer absorption of the constituents in JMT. Secondly, JMT contains a variety of bioactive compounds. There might be noncompetitive antagonism among different compounds, which lead to reduced effectiveness at higher concentration. Besides, some natural products have been found to show opposite effects at different concentrations. For example, emodin promoted proliferation of 3T3-L1 preadipocyte at low concentration and inhibited the proliferation at high concentration in a dose-related manner (Zhang et al., 2002). We have realized that the optional dose span of JMT are in suspense. To solve this problem, the safety and efficacy of JMT in long term administration will be investigated with segmented dose gradients in the near future.

5. Conclusion

Our study investigated the protective effect of JMT against diabetic-induced peripheral neuropathy after different treatment durations at three doses. At the same time, the suppressed Wnt/β-catenin pathway in sciatic nerve of DPN rats were firstly found to be activated by JMT in a time-dependent manner, which probably lead to the up-regulation of MPZ level and the regeneration of peripheral nerve. These findings provide new evidence for JMT as a promising alternative in treatment of DPN, and give us new insight into the potential therapeutic mechanism of JMT.

Authors' contributions

All authors contributed substantially to the work. Xiaochun Liang conceived and designed the experiments. Wei Song and Ying Sun wrote the paper. Ying Sun, Wei Song, Qian Zhang, and Jun Xie analyzed the data. Ying Sun, Chao Wang, and Wei Liu performed the experiments.

Declaration of competing interest

All authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2020.113461>.

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